

THESIS EXECUTION ROADMAP

Complete Guide: June 6 – June 25, 2026

Explainable Multi-Omics Breast Cancer Subtype Classification

Using Consensus Feature Selection, Ensemble Learning & Cross-Omics SHAP Attribution

Student	Rudra (Peash Das Rudra)
University	NUBTK, Khulna
Dataset	TCGA BRCA — 705 patients, 1837 features, 4 omics layers
Defence Date	June 25, 2026
Target	A+ Grade IEEE BIBM 2026 Publication MSc/PhD Profile
Strategy	No CRO implementation — CRO positioned as Future Work only

★ Novel · Publishable · Defensible · A+ Ready ★

1. Strategic Overview

Your supervisor is correct: CRO implementation is not feasible before June 25. This is NOT a weakness — it is reframed strategically. CRO lives entirely in Chapter 7 (Future Work), where it demonstrates mathematical depth and research maturity for MSc/PhD reviewers. Your implemented ML framework IS the contribution.

1.1 Why This Approach is Still Novel & Publication-Ready

- Cross-Omics SHAP Attribution is the scientific finding: you will quantify which omics layer (mRNA / miRNA / CNV / Methylation) drives each breast cancer subtype — this is clinically meaningful and underexplored in the literature
- SMOTE-Inside-CV is a methodological contribution: the majority of published multi-omics papers apply SMOTE before the CV split (information leakage). You will explicitly prevent this.
- 3-Stage Consensus Feature Selection (Variance → ANOVA+MI → RF+XGB) removes single-method bias — a documented gap in feature selection literature
- Stacking Ensemble with Fusion Strategy Comparison answers an open question in multi-omics fusion literature
- NP-Hardness proof (2^{1837} search space, Knapsack reduction) + DC-CRO design in Future Work signals mathematical maturity for any PhD/scholarship committee

1.2 Final Thesis Title

Explainable Multi-Omics Breast Cancer Subtype Classification Using Consensus Feature Selection, Ensemble Learning, and Cross-Omics SHAP Attribution on TCGA BRCA

1.3 Five Research Contributions

#	Contribution	What You Do	Why It is Novel
C 1	3-Stage Consensus Feature Selection	Variance Threshold → ANOVA+MI → RF+XGB consensus ranking	Multi-source agreement eliminates single-method bias
C 2	SMOTE-Inside-CV Pipeline	SMOTE applied ONLY inside training folds — zero leakage	Majority of published papers do this wrong — you will not
C 3	Cross-Omics SHAP Attribution	Quantify mRNA / miRNA / CNV / Methylation contribution per subtype	Key scientific finding — clinically meaningful, underexplored
C 4	Fusion Strategy Comparison	Early fusion (concat) vs Late fusion (per-omics + soft vote) on TCGA BRCA	Answers an open question in multi-omics integration literature
C 5	NP-Hardness + DC-CRO Framework (Future Work)	Mathematical proof of intractability + CRO operator design	Research maturity signal for PhD/funding reviewers

2. Thesis Chapter Structure

CRO does NOT appear as a standalone implementation chapter. It is referenced mathematically in Chapter 2 and positioned as future work in Chapter 7. This is the correct academic framing.

Ch.	Title	Content	Status / Source
1	Introduction	Problem motivation, breast cancer subtypes, multi-omics importance, 5 research objectives, contributions list	<i>Write fresh — Days 11–12</i>
2	Problem Formulation	Search space 2^{1837} , NP-hardness proof via Knapsack reduction, why metaheuristic is required — keep from supervisor presentation	<i>ALREADY DONE — minimal edits</i>
3	Related Work	Multi-omics classification literature, SHAP explainability in genomics, feature selection methods, fusion strategies	<i>Write fresh — Day 12</i>
4	Proposed ML Framework	3-stage feature funnel, SMOTE-inside-CV design, all 8 models, stacking architecture, fusion strategies — your IMPLEMENTED contribution	<i>Days 1–5 work → write Days 13–14</i>
5	Results and Analysis	All 12 figures, model comparison table (8 models × 6 metrics), fusion table, SHAP omics attribution table with biological interpretation	<i>Days 1–5 results → write Day 15</i>
6	Discussion	Interpret SHAP findings biologically (which biomarkers, which omics layer), compare fusion strategies, limitations, why stacking wins	<i>Write Day 16</i>
7	Conclusion and Future Work	Summary of 5 contributions, THEN: DC-CRO design as the natural next step to benchmark against this baseline — CRO shines here	<i>Write last — Day 17</i>

Key: Chapter 2 is ALREADY WRITTEN (from supervisor presentation). Chapter 7 is where CRO lives — as an aspirational future direction, NOT as something you failed to implement.

3. Step-by-Step Execution — Days 1–5

Install all packages FIRST before Day 1 coding: pip install pandas numpy scikit-learn xgboost lightgbm imbalanced-learn shap boruta matplotlib seaborn

DAY 1 · June 6 — Data Pipeline + Stage 1-2 Feature Selection

 **GOAL:** Load all 4 omics layers cleanly, run first two filter stages, end Day 1 with ~300 features ready.

► MORNING — Environment + Data Loading

1. Create project folder: thesis_brca/ with subfolders data/, results/, figures/, models/
2. Install packages: pip install pandas numpy scikit-learn xgboost lightgbm imbalanced-learn shap boruta matplotlib seaborn
3. Load all 4 omics CSVs: mrna.csv, mirna.csv, cnv.csv, methyl.csv + labels.csv
4. Merge on patient barcode (common index column) using `pd.concat(..., axis=1).join(labels)`
5. Verify shape: expected ~705 rows × 1838 columns (1837 features + 1 label)
6. Check missing values per column: `df.isnull().mean()` — drop features >20% missing, impute rest with median
7. Check label distribution: `df["subtype"].value_counts()` — confirm HER2 imbalance (~5.3:1 ratio)
8. Save label distribution bar chart → fig_02_label_dist.png

► AFTERNOON — Stage 1 + Stage 2 Feature Filtering

9. Stage 1 — Variance Threshold: `VarianceThreshold(threshold=0.01).fit_transform(X)`
10. Expected result: 1837 → ~1400 features. Record exact number for thesis.
11. Stage 2A — ANOVA F-test: `SelectKBest(f_classif, k=75)` per omics group
12. Stage 2B — Mutual Information: `SelectKBest(mutual_info_classif, k=75)` per omics group
13. Keep UNION of ANOVA + MI selected features per omics type (OR logic, not AND)
14. Expected result after Stage 2: ~300 features total across all 4 omics
15. Save Stage 2 feature list and omics group membership to CSV → feature_stage2.csv

► EVENING — Label Encoding + CV Setup

16. Encode labels: `LabelEncoder()` — LumA=0, LumB=1, HER2=2, Basal=3, Normal=4
17. Create global CV object: `skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`
18. IMPORTANT: use this SAME skf object for all experiments — ensures consistency
19. Save the feature reduction numbers: 1837 → 1400 → 300 (will extend to 75 tomorrow)

✓ **OUTPUT:** Clean merged dataset (~300 features), label encoded, skf ready, label distribution chart saved as fig_02.

DAY 2 · June 7 — Consensus Feature Selection + All Baseline Models

 **GOAL:** Produce final 50-75 consensus features. Run 5 baseline models with SMOTE-inside-CV. Save results table.

► MORNING — Stage 3: RF + XGB Consensus Ranking

20. Train `RandomForestClassifier(n_estimators=200, random_state=42)` on Stage 2 features
21. Train `XGBClassifier(n_estimators=200, random_state=42)` on Stage 2 features
22. Average importances: `avg_imp = (rf.feature_importances_ + xgb.feature_importances_) / 2`
23. Sort descending, keep top 50–75 features → this is X_final
24. OPTIONAL (if time allows): run Boruta for statistically validated selection — from boruta import BorutaPy
25. Save: top 20 consensus features horizontal bar chart → fig_03_consensus_features.png

26. Record final funnel: 1837 → ~1400 → ~300 → 75 — save funnel figure → fig_01_funnel.png
27. Save X_final and feature names to CSV for SHAP use on Day 4

► AFTERNOON — SMOTE-Inside-CV Baseline Models

28. CRITICAL: use `imblearn.pipeline.Pipeline` NOT `sklearn.pipeline.Pipeline` for SMOTE
29. Pipeline structure: `Pipeline([("smote", SMOTE(random_state=42)), ("clf", model)])`
30. Run 5 baseline models: Logistic Regression, SVM (RBF kernel), KNN (k=5), Naive Bayes, Random Forest
31. Scoring metrics for `cross_validate`: ["accuracy", "f1_macro", "roc_auc_ovr"]
32. Use the global `skf` object from Day 1 for all models — same folds, fair comparison
33. Collect all scores into a pandas DataFrame

► EVENING — Results + Figures

34. Compute mean and std for each metric per model
35. Save results table as CSV → `results_baseline.csv`
36. Plot ROC curves (one-vs-rest multiclass) for all 5 models on one figure → `fig_04_roc_baseline.png`
37. Add Random Forest result to the main comparison table — will be extended on Day 3

✓ **OUTPUT:** 75 final consensus features (X_final), baseline comparison table (5 models × 6 metrics), ROC curve figure saved.

DAY 3 · June 8 — XGBoost + LightGBM Tuning + Stacking Ensemble

🎯 **GOAL:** Train and tune strong models. Build stacking ensemble. Generate confusion matrix. Identify best model.

► MORNING — XGBoost + LightGBM Hyperparameter Tuning

38. XGBoost grid: `n_estimators` [100, 200], `max_depth` [3, 5, 7], `learning_rate` [0.05, 0.1]
39. Use `GridSearchCV` with `cv=skf` and `scoring="f1_macro"` — not accuracy
40. Keep grid small ($2 \times 3 \times 2 = 12$ combinations) — run with `n_jobs=-1` for speed
41. Run same grid search process for LightGBM
42. Print and save `best_params_` for both models

► AFTERNOON — Stacking Ensemble

43. Base learners: RF (`n_estimators=200`) + XGB (best params) + LightGBM (best params)
44. Meta-learner: `LogisticRegression()` — simple, fast, interpretable
45. Use `StackingClassifier` from `sklearn.ensemble` with `cv=5`
46. Wrap in `imblearn Pipeline`: `[("smote", SMOTE(random_state=42)), ("clf", stack)]`
47. Evaluate with `cross_validate` using global `skf` — same as all other models
48. Stacking should outperform all individual models on F1-Macro — if not, check for leakage

► EVENING — Full Metrics + All Figures

49. Compute 6 metrics for ALL models: Accuracy, Precision, Recall, F1-Macro, AUC-ROC, MCC
50. MCC: from `sklearn.metrics` import `matthews_corrcoef` — important for imbalanced classes
51. Build FINAL comprehensive model comparison table: 8 models × 6 metrics — bold the best value per column
52. Save: Confusion matrix heatmap for best model → `fig_06_confusion_best.png`
53. Save: Model comparison bar chart (F1-Macro) → `fig_05_model_comparison.png`
54. Save: Updated ROC curves including XGB, LightGBM, Stacking → `fig_04_roc_all.png`
55. Save all models with `pickle/joblib` for use on Day 4 SHAP analysis

✓ **OUTPUT:** Tuned XGB + LightGBM + Stacking results. Best model identified. Full 8-model comparison table. Confusion matrix. 6 metrics per model. All models saved to disk.

CRITICAL METHODOLOGY SENTENCE (copy this exactly into your thesis): "To prevent information leakage from oversampling, SMOTE was applied exclusively within the training partition of each cross-validation fold, ensuring that synthetic minority-class samples did not influence fold-level validation metrics."

DAY 4 · June 9 — SHAP Analysis — The Core Novelty

 **GOAL:** Generate all 4 SHAP figures. This is the most important day — Cross-Omics SHAP Attribution is what makes this publication-worthy.

► MORNING — Setup + Global SHAP Beeswarm

56. Load best model from disk (from Day 3 saved model)
57. Split data: `X_train, X_test, y_train, y_test = train_test_split(X_final, y, test_size=0.2, stratify=y, random_state=42)`
58. Fit best model on `X_train` — do NOT reuse the cross-validated version, fit fresh for SHAP
59. Create explainer: `shap.TreeExplainer(best_model)`
60. Compute SHAP values: `shap_values = explainer.shap_values(X_test)` → shape (n_samples, n_features, n_classes)
61. Figure 1 — Global SHAP Beeswarm: `shap.summary_plot(shap_values, X_test, plot_type="dot", max_display=20)`
62. Save → `fig_07_shap_beeswarm.png` at dpi=150

► AFTERNOON — Omics Attribution (YOUR KEY FINDING)

63. Load `feature_names` list — ordered same as `X_final` columns
64. Group feature indices by omics prefix (`mrna_`, `mirna_`, `cnv_`, `methy_`) from `feature_stage2.csv` saved on Day 1
65. For each cancer subtype (LumA, LumB, HER2, Basal): compute sum of |SHAP values| per omics group
66. Convert to percentages: each group sum / total sum × 100
67. Result: 4×4 table — 4 omics rows × 4 subtype columns with % contribution
68. Figure 2 — Omics Attribution Grouped Bar: X-axis = omics type, Y-axis = % contribution, grouped bars = subtypes
69. Save → `fig_08_omics_attribution.png`
70. NOTE: This grouped bar chart IS the paper figure for any conference submission

► EVENING — Per-Class SHAP + Patient Waterfall

71. Figure 3 — Per-class SHAP (run once per subtype): `shap.summary_plot(shap_values[:, :, cls_idx], X_test)`
72. Save 4 figures: `fig_09_shap_LumA.png`, `fig_09_shap_LumB.png`, `fig_09_shap_HER2.png`, `fig_09_shap_Basal.png`
73. Figure 4 — Patient Waterfall: select one correctly classified patient per major subtype
74. Use: `explanation = explainer(X_test); shap.plots.waterfall(explanation[patient_idx, :, class_idx])`
75. Save 2–3 waterfall figures: `fig_10_waterfall_p1.png`, etc.
76. Write 3–4 sentences interpreting the omics attribution result biologically — this is the Discussion section core

✓ **OUTPUT:** 4 SHAP figures: Global beeswarm (fig_07), Omics attribution grouped bar (fig_08), Per-class SHAP × 4 (fig_09), Patient waterfall × 2-3 (fig_10). Cross-omics % attribution table filled in.

DAY 5 · June 10 — Fusion Comparison + Final Results Compilation

 **GOAL:** Run early vs late fusion. Compile ALL results. Number and save ALL 12 figures. Experiments are DONE.

► MORNING — Early Fusion (Already Complete)

77. Early fusion = concatenating all omics features → single model (what you have been doing all week)
78. Take the Stacking Ensemble result from Day 3 — this IS the early fusion result
79. Record: F1-Macro and AUC-ROC for early fusion from the Day 3 results table

► AFTERNOON — Late Fusion

80. For each of the 4 omics types: train a separate XGBClassifier on only that omics group features
81. Use the same SMOTE-inside-CV pipeline and global skf for each per-omics model
82. Generate prediction probabilities from each per-omics model on X_test
83. Average probabilities (soft voting): `avg_proba = np.mean([m.predict_proba(X_test) for m in models], axis=0)`
84. Final prediction: `late_pred = np.argmax(avg_proba, axis=1)`
85. Compute F1-Macro and AUC-ROC for late fusion

► EVENING — Final Compilation

86. Build fusion comparison table: Early Fusion vs Late Fusion with F1-Macro and AUC-ROC
87. Save confusion matrix for late fusion model → `fig_12_confusion_late.png`
88. Save fusion comparison bar chart → `fig_11_fusion_comparison.png`
89. Rename and number ALL figures: `fig_01` through `fig_12` — consistent naming
90. Final sanity check: does stacking > all individual models? Is SMOTE never applied to test data?
91. Copy all figures to `figures/` subfolder, all CSVs to `results/` subfolder
92. Write one paragraph summarizing the key SHAP finding — paste into notes for Chapter 6

✓ **OUTPUT:** Fusion comparison table. All 12 figures numbered and saved. Complete results tables (8 models × 6 metrics + 2 fusion strategies + 4×4 omics attribution). EXPERIMENTS COMPLETE.

4. Complete Figures Checklist (12 Figures)

Generate and save ALL figures below. Use exact naming: fig_01_funnel.png through fig_12. These go directly into your thesis Chapter 5 and the conference paper.

#	Day	Figure Name	Filename	How to Generate
1	1	Feature Reduction Funnel	fig_01_funnel.png	4-bar horizontal bar: 1837→1400→300→75 with stage labels
2	1	Label Distribution (subtype counts)	fig_02_label_dist.png	df["subtype"].value_counts().plot(kind="bar")
3	2	Top 20 Consensus Features	fig_03_consensus_features.png	Horizontal bar from avg_importance ranks (RF+XGB average)
4	2-3	ROC Curves — All Models (OvR)	fig_04_roc_all.png	sklearn roc_curve, one-vs-rest per model on same axes
5	3	Model Comparison Bar (F1-Macro)	fig_05_model_comparison.png	Grouped bar from results DataFrame — all 8 models
6	3	Confusion Matrix — Best Model	fig_06_confusion_best.png	ConfusionMatrixDisplay.from_predictions() — 5×5 heatmap
7	4	SHAP Global Beeswarm	fig_07_shap_beeswarm.png	shap.summary_plot(plot_type="dot", max_display=20)
8	4	Omics Attribution Grouped Bar	fig_08_omics_attribution.png	Matplotlib grouped bar: 4 omics × 4 subtypes × % contribution
9	4	Per-Class SHAP (×4 subtypes)	fig_09_shap_classes.png	shap.summary_plot per class index — one per LumA/LumB/HER2/Basal
10	4	SHAP Waterfall (2-3 patients)	fig_10_waterfall_patient.png	shap.plots.waterfall(explanation[idx, :, class_idx])
11	5	Fusion Comparison Bar Chart	fig_11_fusion_comparison.png	Simple 2-bar chart: Early vs Late Fusion F1-Macro and AUC-ROC
12	5	Confusion Matrix — Late Fusion	fig_12_confusion_late.png	Same as Fig 6 but for late fusion predictions

5. Results Tables (Fill In Your Values)

5.1 Main Model Comparison Table

Fill in -- with your computed values after Days 2-3. Bold the best value in each column.

Model	Accuracy	F1-Macro	Precision	Recall	AUC-ROC	MCC
Logistic Regression	--	--	--	--	--	--
SVM (RBF)	--	--	--	--	--	--
KNN (k=5)	--	--	--	--	--	--
Naive Bayes	--	--	--	--	--	--
Random Forest	--	--	--	--	--	--
XGBoost (tuned)	--	--	--	--	--	--
LightGBM (tuned)	--	--	--	--	--	--
Stacking Ensemble	--	--	--	--	--	--

5.2 Fusion Strategy Comparison

Fusion Strategy	F1-Macro	AUC-ROC	Model Used
Early Fusion (concatenate all omics → 1 model)	--	--	Stacking Ensemble (RF + XGB + LightGBM)
Late Fusion (per-omics model → soft vote)	--	--	XGBoost × 4 (one per omics type)

5.3 Omics Attribution Table (from SHAP — Day 4)

This is the KEY SCIENTIFIC FINDING. The omics attribution percentages from SHAP go here. Interpret biologically: which omics layer drives each subtype?

Omics Layer	LumA (%)	LumB (%)	HER2 (%)	Basal (%)
mRNA Expression	--	--	--	--
miRNA	--	--	--	--
Copy Number Variation (CNV)	--	--	--	--
DNA Methylation	--	--	--	--
TOTAL	100%	100%	100%	100%

6. Defense Slide Plan — 8 Slides for A+

These 8 slides, in this order, are specifically designed to impress both NUBTK faculty AND future MSc/PhD reviewers. Practice 3+ times. Prepare answers for the Q&A section below.

#	Slide Title	Content	Why It Impresses
1	Title + Problem Statement	Research title, TCGA BRCA dataset stats (705 patients, 1837 features, 4 omics), subtype distribution chart (fig_02)	Clear scope, professional opening — dataset complexity signals rigor
2	NP-Hardness (1 slide only)	2^{1837} search space visual, Knapsack reduction mapping (from Chapter 2), conclusion: metaheuristic required — CRO designed as future solution	Mathematical depth — nobody else at NUBTK has this slide. PhD signal.
3	Pipeline Architecture Diagram	Clean flow: Raw 4-omics Data → 3-Stage Feature Funnel → SMOTE-Inside-CV → 8 Models → SHAP → Results	Visual overview — makes the entire methodology immediately clear
4	Feature Reduction Funnel	Bar chart: 1837 → 1400 → 300 → 75 with stage names and methods (fig_01). Show the consensus ranking logic.	Concrete, visual, shows systematic thinking and stage-by-stage justification
5	Model Comparison Results	Full table: 8 models × 6 metrics. Highlight best row in green. Point out SMOTE-inside-CV and why it matters.	Thorough evaluation impresses technical faculty. MCC shows class-imbalance awareness.
6	SHAP Beeswarm — Global	Show fig_07. Say: "These are the top 20 biomarkers our model uses." Name the top 3 features and their biological role.	THE WOW SLIDE. Nobody else at NUBTK has SHAP. Explainability = clinical relevance.
7	Omics Attribution Finding	Show fig_08. Say: "We find that mRNA dominates HER2 classification, while Methylation is more prominent in Basal subtypes." (use your actual numbers)	KEY SCIENTIFIC FINDING. This is the conference paper result. Clinically meaningful insight.
8	Future Work: DC-CRO	Show the DC-CRO operator design diagram from Chapter 3. Say: "This ML baseline will be used to benchmark the CRO-based optimizer in the next phase."	CRO shown as forward-looking, not missing. PhD/MSc committees see ambition + feasibility.

6.1 Q&A Preparation — Likely Questions

Prepare SHORT, CONFIDENT answers (30–60 seconds each). Do not over-explain.

Q: Why did you not implement CRO?

A: CRO is mathematically complex and computationally expensive to implement correctly in the available time. I designed the full DC-CRO framework in Chapter 3 and the ML system here is specifically built to serve as the benchmark baseline when CRO is implemented in the next phase.

Q: Why SMOTE inside CV?

A: Applying SMOTE before the cross-validation split causes synthetic samples from training to appear in validation folds — this inflates reported performance. I applied SMOTE only inside each training fold to ensure validation metrics reflect true generalization.

Q: What is the key finding?

A: The key finding is in the SHAP Omics Attribution: we quantify for the first time on TCGA BRCA how much each omics layer — mRNA, miRNA, CNV, Methylation — contributes to classifying each breast cancer subtype. For example, [state your actual number] of HER2 classification is driven by mRNA expression.

Q: Why consensus feature selection?

A: Using a single feature selection method can introduce bias specific to that method. By requiring consensus between Random Forest and XGBoost importance rankings (after ANOVA+MI pre-filtering), we select features that are robust across multiple selection criteria.

Q: Why stacking ensemble?

A: Stacking combines the strengths of diverse base learners. RF, XGBoost, and LightGBM each capture different patterns in high-dimensional omics data. The meta-learner learns how to weight their predictions optimally, consistently outperforming any single model.

7. Complete Timeline — June 6 to June 25

Date	Day	Task	Deliverable
June 6	1	Data pipeline, preprocessing, Stage 1-2 feature selection	Clean dataset ~300 features, label dist chart (fig_02)
June 7	2	Stage 3 consensus ranking, all 5 baseline models with SMOTE-CV	75 final features, feature funnel (fig_01, fig_03), baseline table
June 8	3	XGBoost, LightGBM tuning, Stacking Ensemble, 6 metrics	Full 8-model comparison table, confusion matrix, ROC curves
June 9	4	ALL 4 SHAP analyses — beeswarm, omics attribution, per-class, waterfall	4 SHAP figures (fig_07–10), omics attribution % table filled
June 10	5	Early vs Late fusion comparison, final compilation	All 12 figures numbered, all tables complete — EXPERIMENTS DONE
June 11–15	Writing	Write thesis Chapters 1, 3 (related work), 4, 5, 6, 7 (Chs 2–3 done)	Complete thesis draft
June 16–18	Revision	Supervisor review, revisions, formatting, references, abstract	Polished thesis document
June 19–22	Defense Prep	Build 8-slide presentation, practice 3+ times, prepare Q&A answers	Defense slides ready, answers rehearsed
June 23–24	Final Polish	Final checks, print thesis if needed, rest and mental preparation	100% ready
June 25	DEFENSE	Present with confidence — you have the best prepared thesis in the room	A+ Grade

PUBLICATION TARGET — IEEE BIBM 2026

Submission Window: August–September 2026 | Submit within 2 months of defense

Paper title: "Explainable Multi-Omics Breast Cancer Subtype Classification via Consensus Feature Selection and Cross-Omics SHAP Attribution"

8. Chapter 7 — Future Work: DC-CRO Framework

This section shows you EXACTLY what to write in Chapter 7 to present CRO as a sophisticated future direction. CRO is not missing — it is the natural next phase.

8.1 How to Position CRO in Chapter 7

The following structure ensures CRO appears as a deliberate and mathematically rigorous future direction — not as something you failed to implement:

1. State clearly: "The ML framework developed in this thesis serves as a baseline benchmark for evaluating optimization-based approaches."
2. Reference Chapter 2: "As established in Chapter 2, the feature selection problem over 2^{1837} combinations is NP-hard via reduction to the 0/1 Knapsack problem, motivating the use of metaheuristic search."
3. Introduce DC-CRO: "A Dual-Core Chemical Reaction Optimization (DC-CRO) framework has been designed to address this NP-hardness, incorporating four reaction operators adapted for binary omics feature selection."
4. State the benchmark plan: "Future work will implement DC-CRO and compare it against the stacking ensemble baseline established here, using identical evaluation metrics (F1-Macro, AUC-ROC, MCC) to quantify the benefit of optimization-guided feature selection."
5. Mention expected outcome: "Based on the established mathematical properties of CRO and similar metaheuristics applied in genomics, DC-CRO is expected to identify a more compact feature subset with equal or superior classification performance."

8.2 DC-CRO Framework (Already Designed — Reference from Chapter 3)

In Chapter 7, reference the design from Chapter 3 and add:

- DC-CRO Objective Function: minimize feature count while maximizing F1-Macro $\rightarrow f(S) = F1(S) - \lambda \cdot |S|/n$
- Four reaction operators already adapted: Decomposition, Synthesis, Inter-molecular Collision, On-wall Ineffective Collision
- Energy conservation mechanism maps to feature evaluation cost in the omics search space
- Implementation plan: estimated 2–3 months for full CRO implementation and benchmarking
- This positions the thesis as Phase 1 of a multi-phase research program — ideal for MSc/PhD proposals

KEY FRAMING: Say "this thesis establishes the performance baseline" not "we could not implement CRO." The distinction signals research maturity.

9. Package Reference

Package	Version	Used For
pandas, numpy	latest	Data loading, manipulation, array operations, results tables
scikit-learn	≥ 1.3	Feature selection, baseline models, StratifiedKFold, all metrics including MCC
xgboost	≥ 1.7	XGBoost classifier, feature importance, SHAP TreeExplainer compatibility
lightgbm	≥ 3.3	LightGBM classifier — fast, excellent on high-dimensional omics data
imbalanced-learn	≥ 0.11	SMOTE inside Pipeline — CRITICAL for correct leakage-free evaluation
shap	≥ 0.44	TreeExplainer, beeswarm plot, waterfall plot, summary_plot per class
boruta	latest	Optional Stage 3: statistically validated wrapper around Random Forest
matplotlib, seaborn	latest	All figures: ROC curves, confusion matrix heatmaps, bar charts, SHAP plots

Install command: `pip install pandas numpy scikit-learn xgboost lightgbm imbalanced-learn shap boruta matplotlib seaborn`

START DAY 1 — JUNE 6. First action: load data and verify merge shape.

Everything else builds from that first clean merge. You have the best strategy. Execute.